

Bryan Ramirez

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SUMMARY

UC San Diego Cognitive Science student specializing in Machine Learning & Neurocomputation with a minor in Economics. My background combines Python/Pandas data analysis, SQL fundamentals, Excel/Google Sheets experience, statistical coursework, AI-assisted workflows, and clear technical documentation. I bring a careful, practical approach to cleaning data, checking outputs, finding patterns, organizing results, and explaining analytical findings in plain language.

EDUCATION

University of California, San Diego — B.S. Cognitive Science — Specialization in Machine Learning & Neurocomputation — Expected June 2026 — Minor in Economics

Relevant Coursework: Python, Data Science in Practice, Machine Learning, Neural Networks and Deep Learning, Modeling and Data Analysis, Statistical Analysis, Probability, Large Language Models, Microeconomics

TECHNICAL PROJECTS

Air Quality and Health Impact Analysis | Python, Pandas, scikit-learn

- Analyzed a 5,811-record dataset connecting air quality, weather variables, and health impact outcomes.
- Cleaned and explored environmental variables, reviewed missing values, checked class imbalance, and documented relationships between predictors and outcomes.
- Compared regression models including Linear Regression, Lasso, Polynomial Regression, and Principal Component Regression using cross-validation and held-out testing.
- Built classification workflows for imbalanced health impact classes using Logistic Regression and KNN, documenting accuracy, balanced accuracy, macro F1-score, and limitations.

Object Detection Benchmark | Python, PyTorch, YOLOv8, Faster R-CNN, DETR |

github.com/BryanR-17/Object-Detection-Benchmark

- Built a Python benchmarking workflow comparing three object detection models on 300 Pascal VOC images.
- Tracked runtime, average inference time, and frames per second, then organized outputs into tables and visual comparisons.
- Documented performance differences clearly to support review of model tradeoffs and practical inference behavior.

Machine Learning Model Evaluation Study | Python, Scikit-learn

- Built and evaluated Logistic Regression, Random Forest, and MLP models using validation splits and repeated trials.
- Compared results across runs, documented performance differences, and reviewed outputs for consistency and reliability.
- Practiced structured testing, troubleshooting, and clear reporting of analytical findings.

Smartphone Usage and Mental Health Analysis | Python, Pandas, scikit-learn

- Cleaned, transformed, and analyzed behavioral datasets related to smartphone usage and mental health indicators.
- Applied regression and clustering models to identify patterns, compare relationships, and summarize findings.
- Organized analysis steps into a repeatable workflow for reliable interpretation, reporting, and troubleshooting.

EXPERIENCE

Lumper — TW Services | La Mirada, CA | Mar 2021 – Present

- Maintain efficiency in high-volume operations by following structured workflows, coordinating task completions and supporting accurate, on time execution in a fast-paced environment
- Communicate with team members to resolve workflow issues, adapt to changing priorities, and keep operational work moving reliably
- Follow established procedures, identify issues quickly, and maintain consistency across routine tasks in a fast-paced environment

SKILLS

- Data Analysis: data cleaning, validation, EDA, trend identification, basic statistical analysis, quality checks, reporting-ready analysis
- Tools: Python, Pandas, NumPy, SQL fundamentals, Excel, Google Sheets, scikit-learn, Jupyter, GitHub, VS Code, CSV/spreadsheet workflows
- Reporting and Visualization: charts, tables, analytical summaries, dashboard concepts, KPI-style reporting, findings review, workflow documentation
- AI-Assisted Workflow: using ChatGPT and similar tools to reason through code, speed up documentation, troubleshoot errors, and review analytical logic
- Technical: JavaScript, HTML, CSS, R, MATLAB, PyTorch, TensorFlow, SciPy
- Strengths: attention to detail, analytical thinking, plain-English communication, reliability, collaboration, fast learning, follow-through

EXTRACURRICULARS

- Society of Hispanic Professional Engineers (SHPE): Participate in technical workshops and mentorship programs
- Cognitive Science Student Association (CSSA): Engage in discussion and events related to AI, cognition, and technology